



**HWA CHONG INSTITUTION (COLLEGE SECTION)**  
**2018 JC2 9744 H2 BIOLOGY**  
**PRELIMINARY EXAMINATION PAPER 4 MARK SCHEME**

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**QUESTION 1**

- (a) Describe the steps to be taken for determining the exact volume of **H** required to fill the column to the 10 cm<sup>3</sup> mark. [2]
- Ref. to fixed volume of H obtained
  - Ref. to initial volume minus final volume to obtain volume of H
- (b) Record the volume of **H** added into the column. [1]
- Ref. to acceptable range of results
- (c) Complete Fig. 1.1. [3]
- Ref. to dilutions showing appropriate volumes of H
  - Ref. to dilutions showing appropriate volumes of W
  - Ref. to appropriate final concentrations
- (d) Suggest how simple dilution can allow for more accurate results to be obtained. [2]
- Ref. to errors that occur between dilutions of serial dilution will not be carried forward
  - Ref. to simple dilution allowing values close to true values to be obtained
- (e) Record your results in a suitable table in the space below. [3]
- Correct column headings with units
  - Show final correct concentrations obtained
  - Show correct trend observed
- (f) Record the minimum percentage of H<sub>2</sub>O<sub>2</sub> present in the eluent. [1]
- Ref. to acceptable range of results
- (g) Calculate the relative activity of catalase from watermelon. [2]
- Show your working clearly.
- Ref. to proper presentation of workings
  - Ref. to obtaining rate by dividing by incubation time
- (h) Other than replicates and repeats, describe and explain **one** modification to the method that will increase confidence in the results obtained. [2]
- Ref. to keeping surface area is constant
  - Ref. to use of thermostatically controlled water bath
  - Ref. to replacing watermelon with distilled water
- (i) Use the grid to display the results shown in Table 1.1 in an appropriate form. [3]
- Graph showing sensible scale including use of correct axes labels;;
  - Correct plotting of the bar graph
  - Correct plotting of the error bars



(j) State what the standard deviation,  $s$ , shows about the results of this investigation. [1]

- Ref. to  $s$  shows the spread of data the reliability of the results

(k) The student carried out a statistical test to find out if the difference in the rate of reaction between the immobilised and free enzyme was significant.

The results were significant at  $p < 0.05$ . Explain what this means. [2]

- Ref. to less than 5% possibility
- Ref. to that the difference is not due to chance

[Total: 22]

## **QUESTION 2**

(a)

- Ref to catalysis of hydrogen peroxide being dependent on the temperature
- Ref to concept on non-competitive inhibition
- Equilibrate separately solutions of catalase,  $\text{H}_2\text{O}_2$  and  $\text{CuSO}_4$
- Ensure the reaction occurs in an air tight system
- Measure the volume of oxygen produced with and without presence of inhibitor
- Repeat at  $> 5$  different temperatures
- Determine % difference in max rate of reaction
- Illustrate results using a graph
- Wear goggles and gloves to protect against irritants
- AVP

(b) Based on your answer in 2(a),

(i) suggest a variable that was not taken into consideration in your plan when studying the catalase activity in *Ae. aegypti* female mosquitoes. [1]

- temperature tolerance range mosquitoes
- may not reflect in vivo conditions
- AVP

(ii) identify and explain one possible biological impact of using the copper spray to combat mosquito breeding in the natural environment. [2]

- Ref to negative impact on other organisms
- AVP

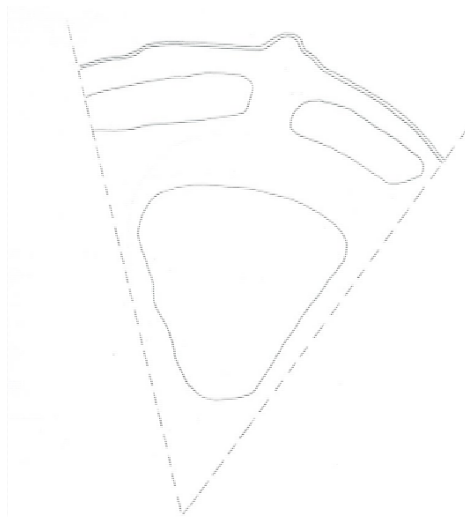
[Total: 14]

### QUESTION 3

- (a) (i) You are required to use a sharp pencil for drawing.

Use the space provided to draw a plan diagram of part of the stem on slide **S1**, as shown in the shaded area of Fig. 3.1. A plan diagram only shows the arrangement of the different types of tissues. Individual cells must **not** be drawn in plan diagrams. [3]

- Drawing quality of appropriate size with clear continuous lines
- Correct section drawn
- Draw the different tissue layers in correct proportion



- (ii) Observe the epidermal layer of the transverse section of the *Cucurbita pepo* stem in slide **S1**.

State which objective lens you have decided to use.

[1]

40x

- (iii) Using the eyepiece graticule fitted in the eyepiece lens of your microscope, and the stage micrometer, calculate the actual thickness, in  $\mu\text{m}$ , of the epidermal layer observed in slide **S1**.

Show your working clearly.

[3]

- Shows division of stage micrometer measurement by number of eyepiece graticule divisions with correct working
- Shows measurement of width of epidermal cell from slide in eyepiece graticule divisions
- Correct calculation/working AND final answer in  $\mu\text{m}$

- (b) Slide **S2** contains two stained transverse sections of the *Cucurbita pepo* root. The slide shows actively dividing cells in the zone of cell division. Make large labelled drawings of **two** cells showing chromosomes undergoing **different** stages of mitosis. [4]

- Quality drawing of appropriate size that includes clear continuous lines with no shading AND of two cells only
- Correct drawing of cell wall represented with double lines
- Correct drawing of chromosomes AND nuclear envelope matching identified stage of mitosis observed
- At least 3 correct labels: cell wall, cell membrane, chromosomes and cytoplasm

- (c) (i) Count the total number of cells in a field of view.

[1]

150 to 450 cells in one FOV

- (ii) Count the number of cells that are in each stage of mitosis as observed in the same field of view seen in (c)(i).

Record your results in a suitable table in the space below.

[3]

- Represents data in the form of a table
- Appropriate table headings 'stage of mitosis' and 'number of cells'
- Record reflects prophase with highest number of cells counted

- (iii) Calculate the mitotic index. Show your workings clearly.

[2]

- Shows working
- Shows correct value and recorded to appropriate sf

- (d) Describe two ways in which the cells observed in Fig. 3.2 differ from those in slide S2. [2]

- Cells in Fig. 3.2 are rectangular in shape compared to slide S2 cells which are circular / Cell in Fig. 3.1 are longer than the cells drawn in slide S2
- Cells in Fig. 3.2 are less actively dividing / undergoing mitosis than cells in slide S2
- Cells in Fig. 3.2 have a large area of cytoplasm: nucleus compared to cells in slide S2

[Total: 19]